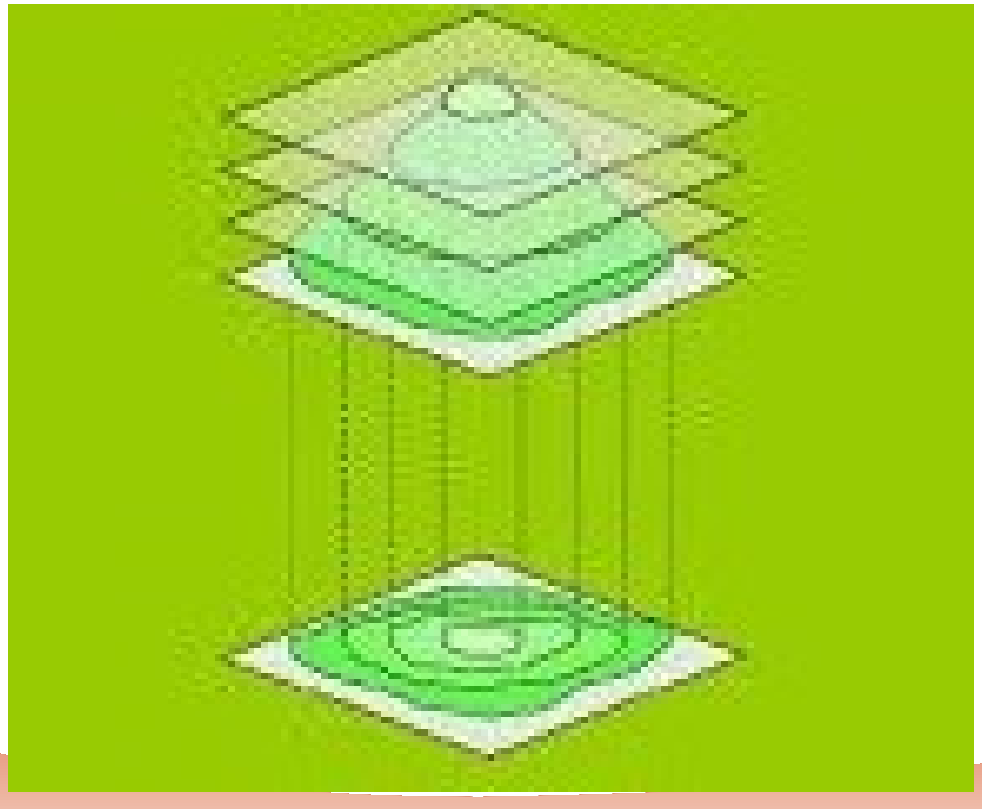
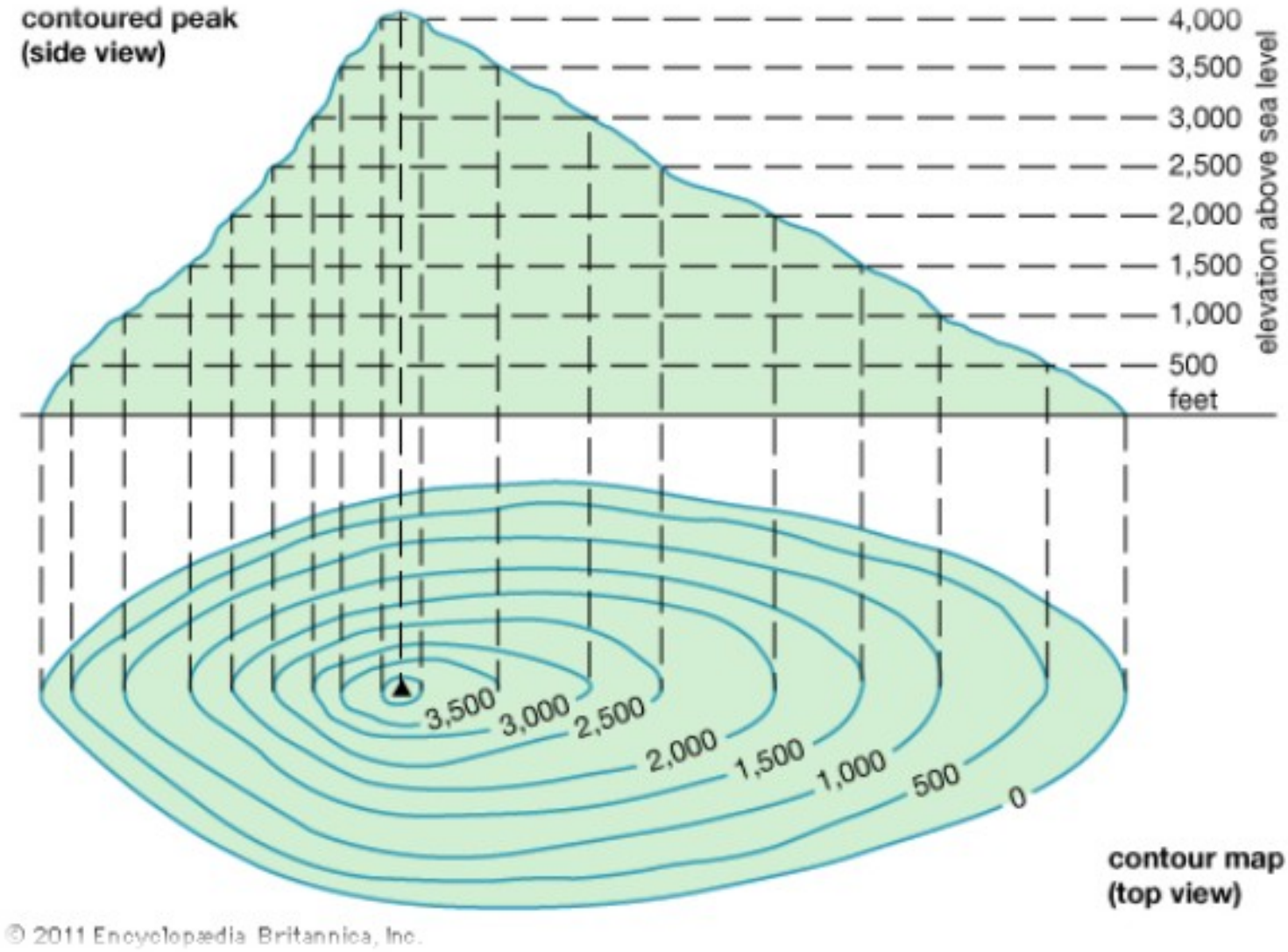
CONTOURING

“An imaginary line on ground joining the points of equal elevation above the assumed datum is called a contour”.

- **It is a plan projection of the plane passing through the points of equal height on the surface of the Earth.**



- **The elevations and depressions of the surface of the ground are shown on a map by means of contour lines.**



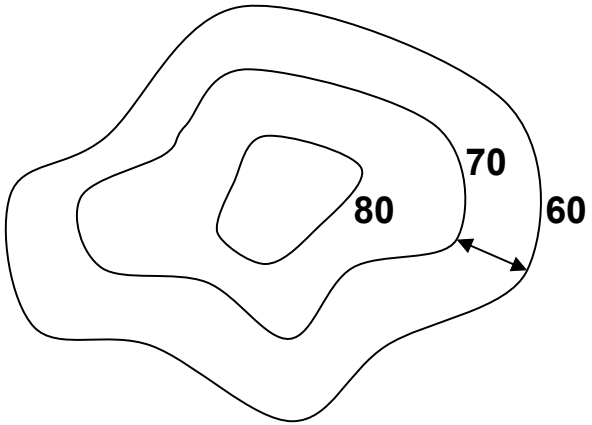
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- **Contour interval:**

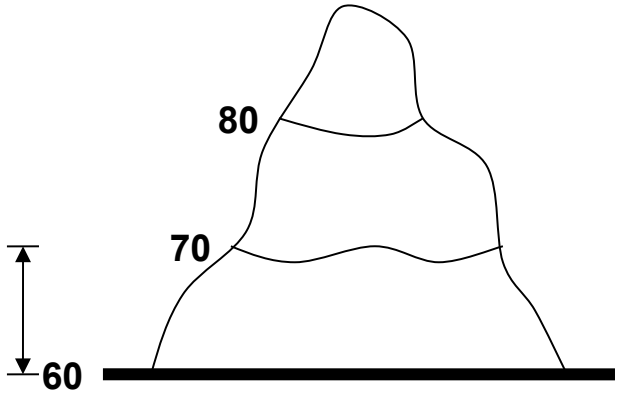
  The vertical distance between any two consecutive contours is called the *contour interval*.

- **Horizontal equivalent:**

  The least horizontal distance between any two consecutive contours is known as the *horizontal equivalent*.



Horz equivalent



Contour interval

For a given contour interval, the horizonal equivalent depends upon the followings:-

a. Nature of the ground.

b. Purpose and extent of the survey.

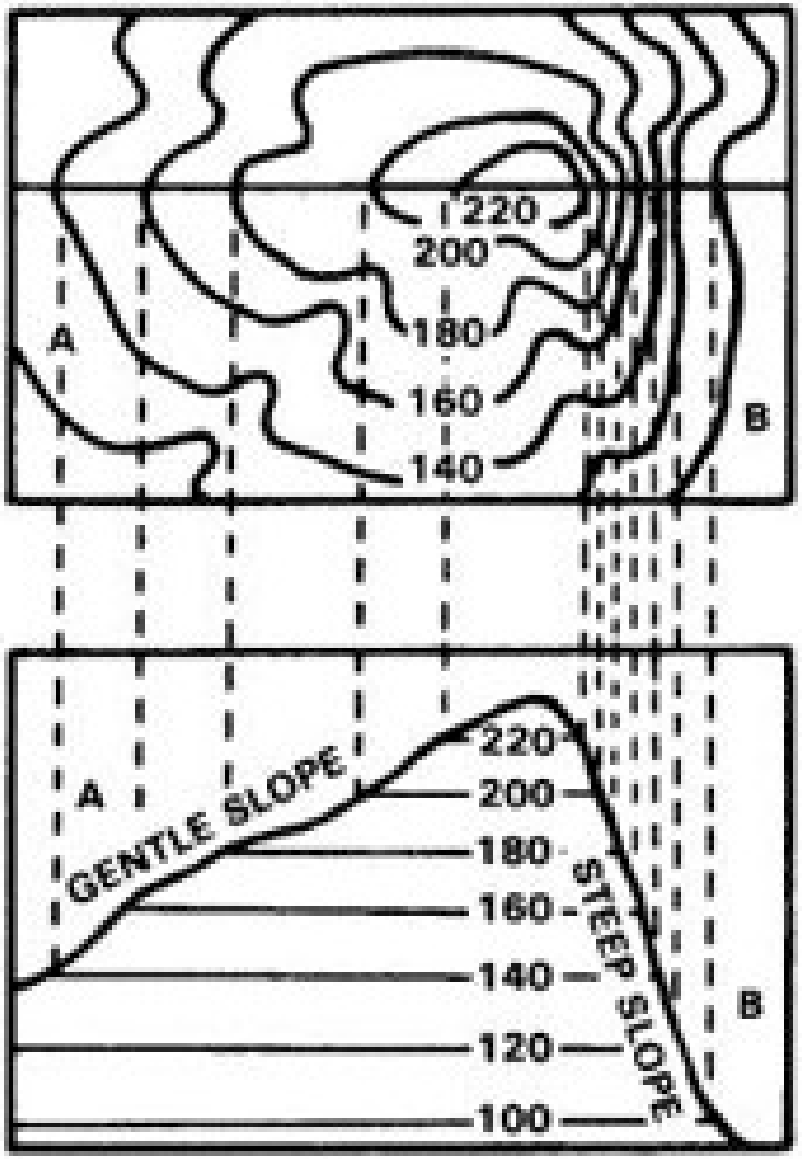
c. Scale of the map.

d. Time and expense required in its determination.

Characteristics of Contour Lines

1. Contour lines run close together near the top of the hill, representing very steep ground, and wide apart at the foot of a hill indicating flat ground.

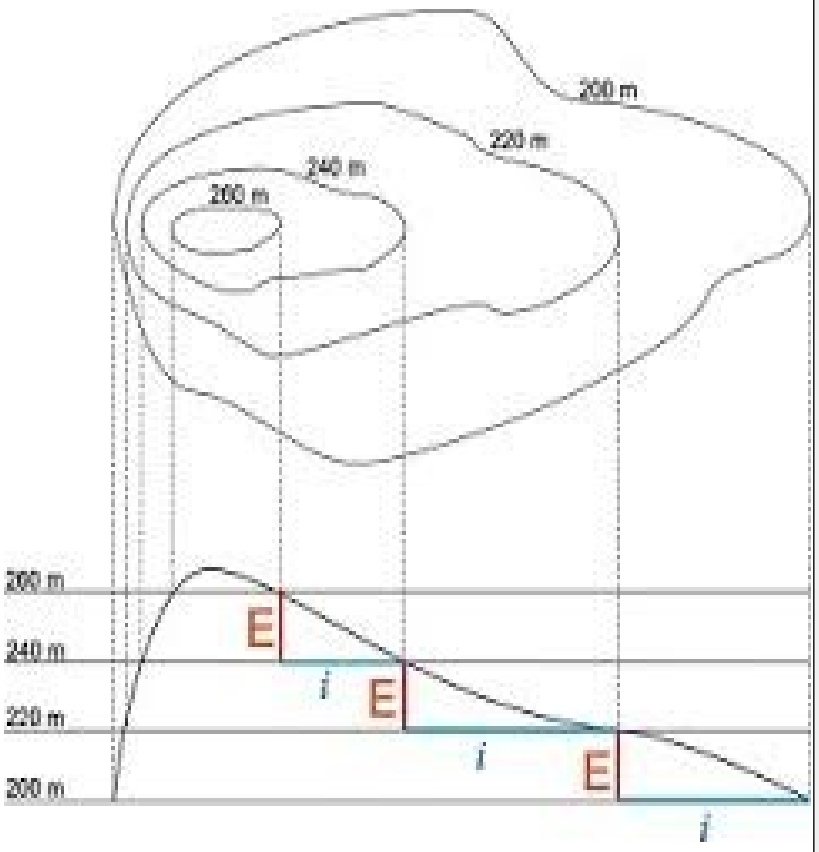
2. A uniform slope is indicated when contour lines are uniformly spaced, while a plane surface is indicated when they are straight and equally spaced.



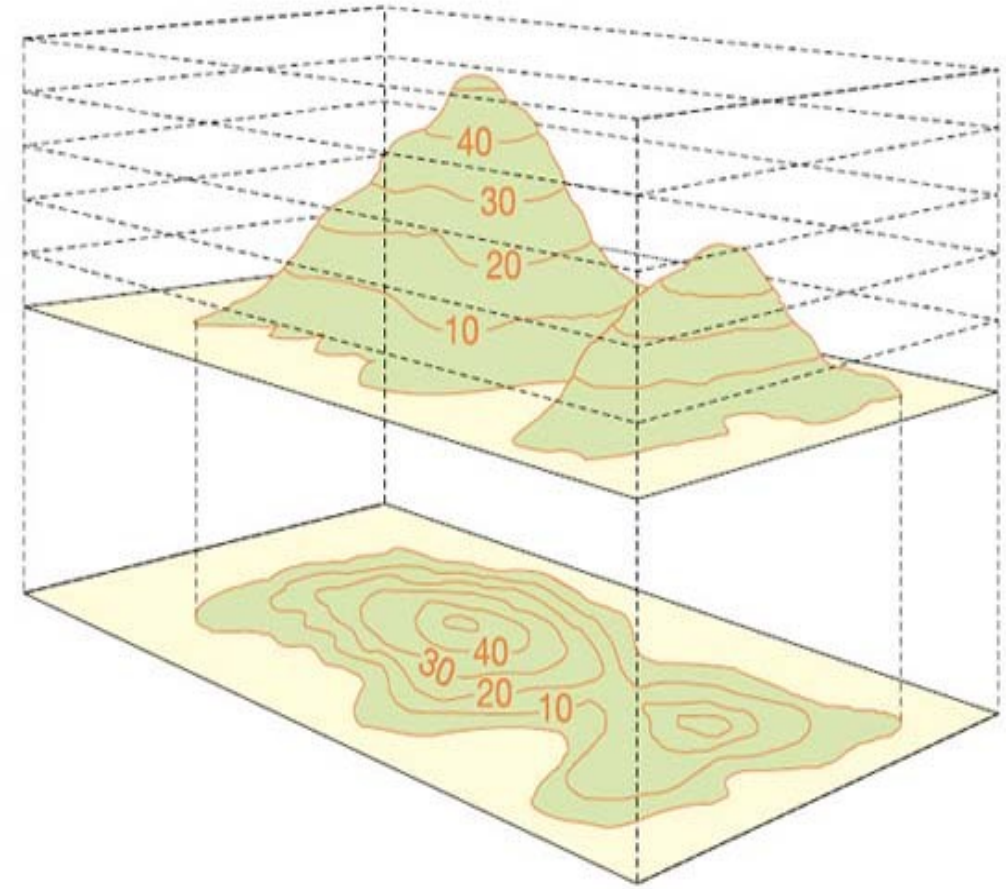
3. Contour lines cannot merge or cross one another on the map, except in the case of an overhanging cliff. A vertical cliff is indicated when several contours coincide, the horizontal equivalent being zero.

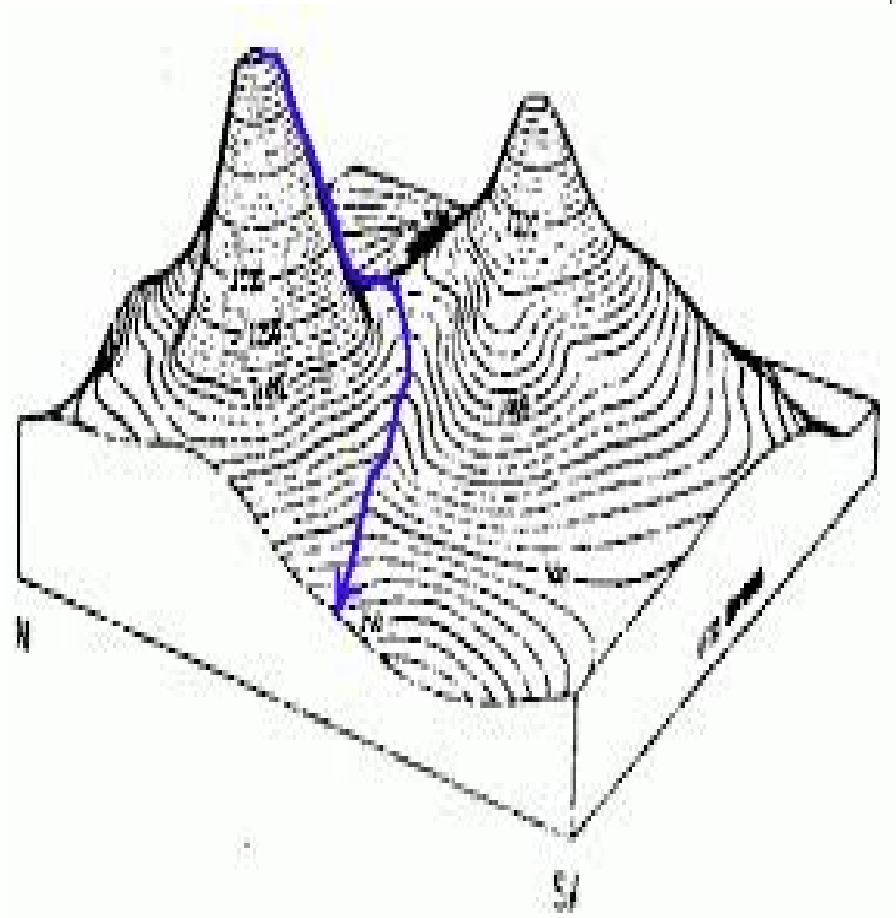
4. Contour lines cannot end anywhere, but close on themselves either within or outside the limits of the map.

5. Contours do not have sharp turnings.



6. A series of closed contours on the map indicates a pond/depression or a hill/summit, according as the lower or higher values are inside them. Depressions between summits are called saddles.





Use of CONTOURS

1. By Inspection of a contour map, information regarding the general character of the tract of the country is obtained, whether it is flat, undulating or mountainous, etc.

2. The most economical or suitable site for engineering works such as a reservoir, canal, sewer, road, or railway may be approximately selected.

3. Quantities of earthwork may be computed from the contour maps.

4. Contours may be used to determine the area of the drainage basin and the capacity of the reservoir.

5. The sections may be easily drawn in any direction from the contours.

6. Intervisibility of two given points can be ascertained from the map.

7. A route of a given grade line can be traced on the map.

8. To decide the best positions of guns, the line of march and camping grounds by the army commanders during wars.

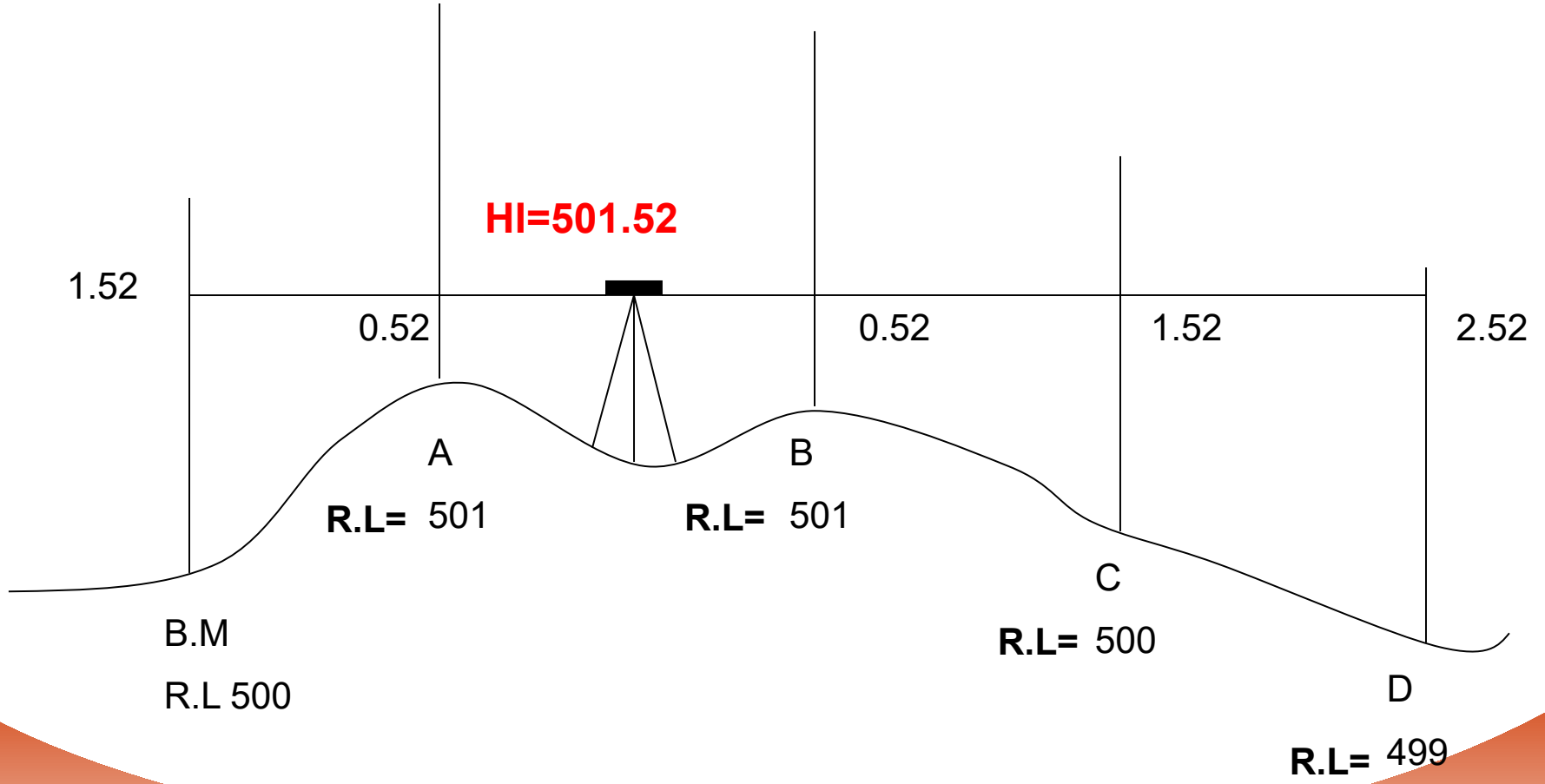
Locating Contours

The various methods of locating contours may be classified as follows:

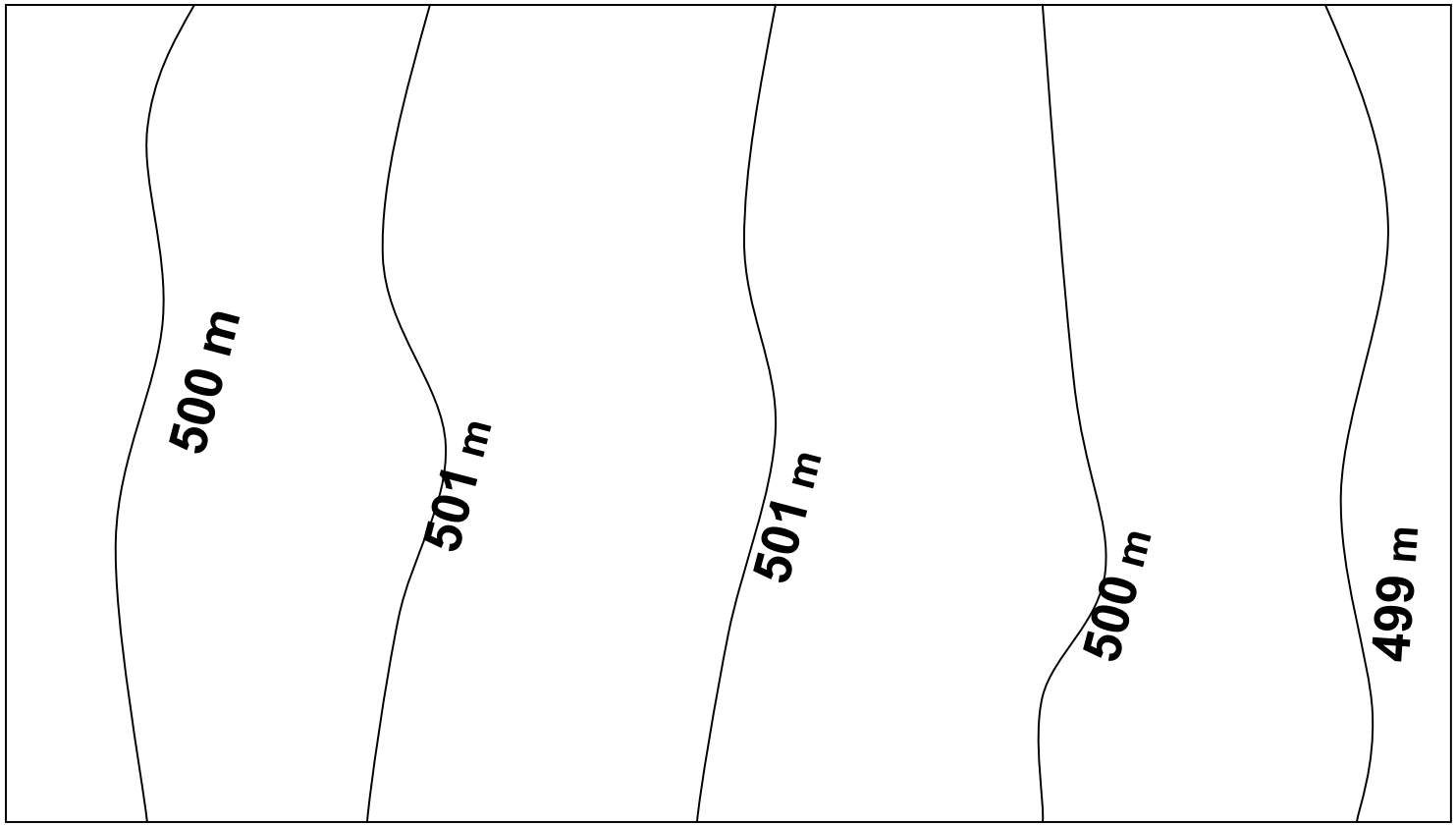
1. Direct Method
2. Indirect Method

Direct Method:

In this method, contours to be plotted are actually traced out in the field. This method is comparatively slow and is generally not adopted on large surveys unless a superior accuracy is demanded. It is suitable for contouring of small areas where better accuracy is required.


HI=501.52
1.52
0.52
0.52
1.52
2.52
A
R.L= 501
B
R.L= 501
C
R.L= 500
D
R.L= 499
B.M
R.L 500

Contour Lines:



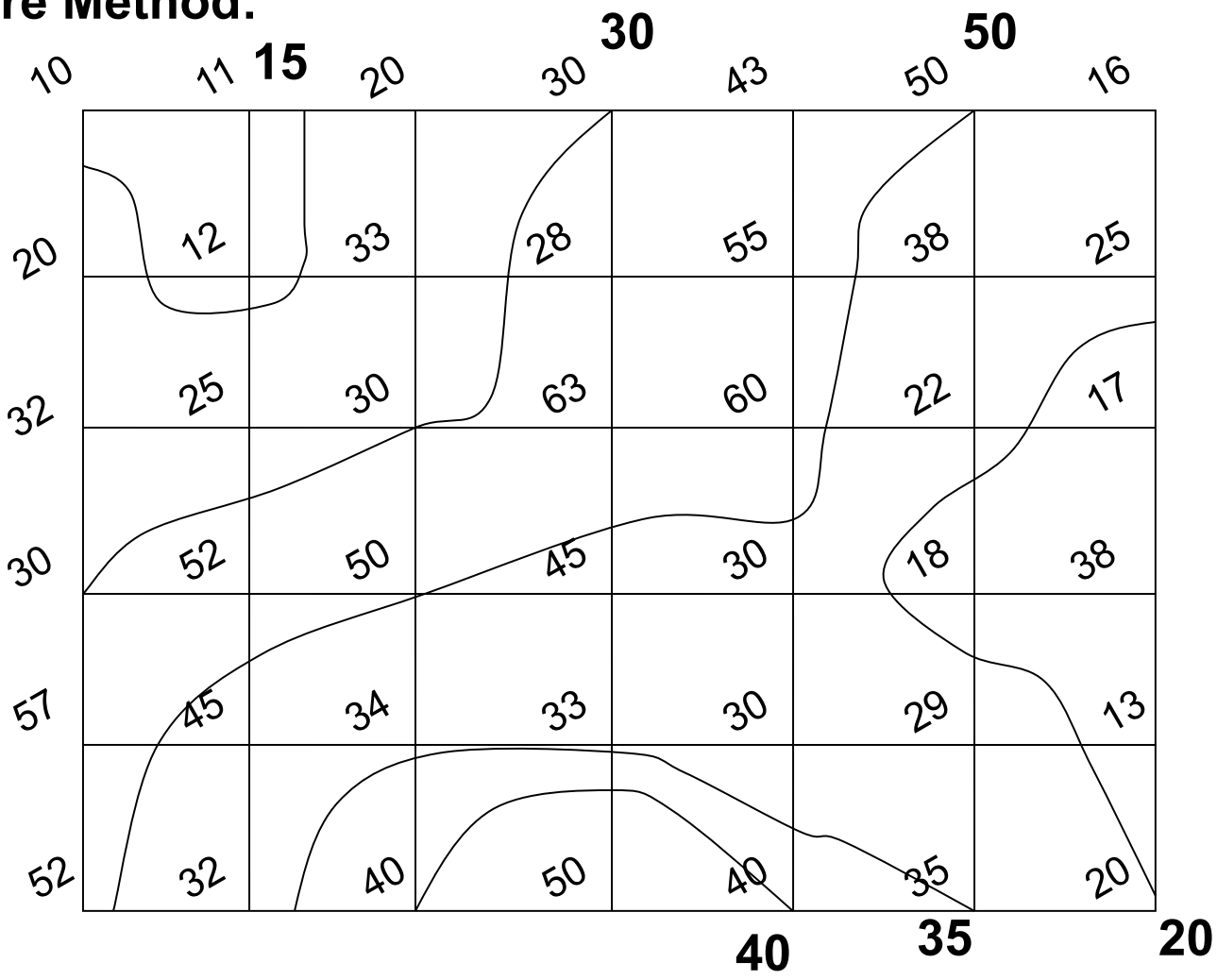
2. Indirect Methods:

Indirect method of contouring is commonly employed in small scale surveys of extensive areas. This method is cheaper, quicker and less tedious as compared with direct method of contouring. Indirect method of contouring can be employed in three different ways detailed below:

a. By Squares Method
b. By Cross Sections Method
c. By Radial Lines Method

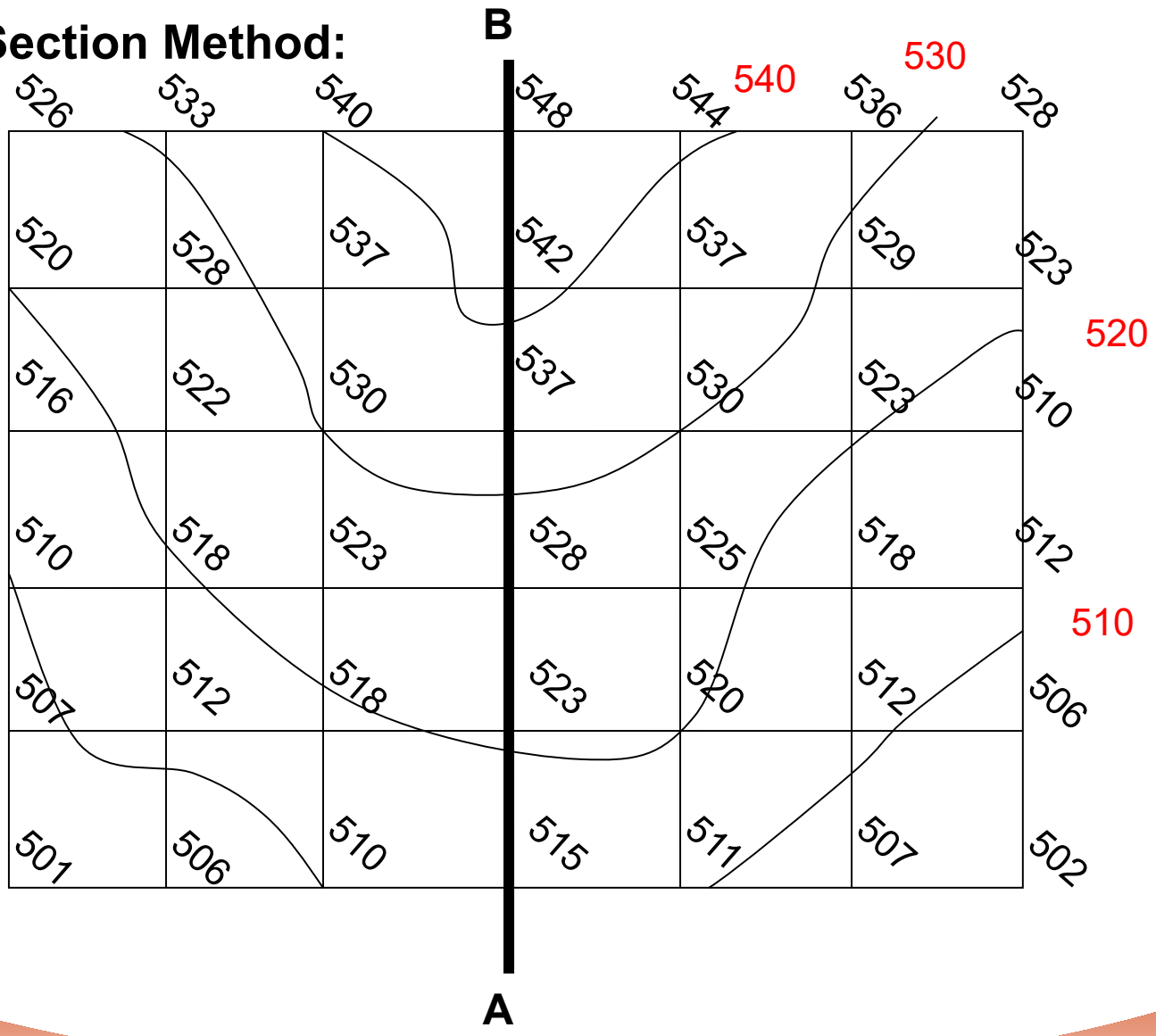
- **By Square Method:** This method is suitable if the area is not very extensive. In this method the area is divided into a series of squares, and the corners of the squares are marked with pegs. The size of the squares varies from 5m to 20m side, depending upon the nature of the ground and the contour interval. The size of the squares need not to be the same throughout. Each separate square is pegged out. The elevations of the ground at the corners of the squares are determined with a level. The system of the square is plotted and elevation of each corner is written next to it. The contour lines are then interpolated in the usual way.

By Square Method:



- **By Cross Section Method:** This method is most commonly used in a route surveys of a road, railway or canal alignment. Cross-sections are run transverse to the centre line of a road, railway, or canal, and the points of change in slope are located. The cross-section line may be inclined at an angle to the centre line if necessary. The spacing of the cross-sections depends upon the character of the ground. It is usually 20m in a hilly country, and 100m in a flat country. The reduced levels of the various points along the section lines are plotted on the plan and the contours are then interpolated.

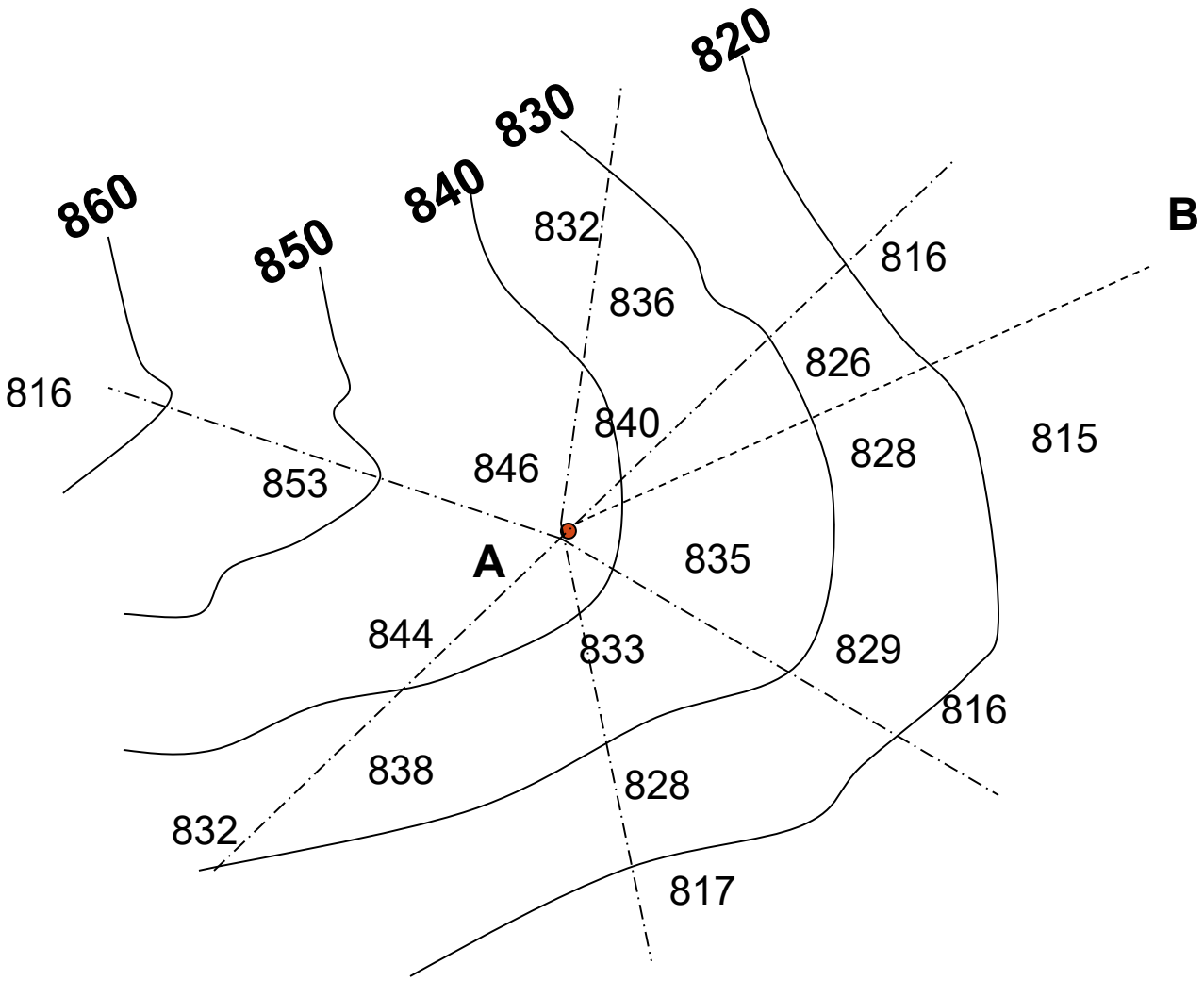
By Cross Section Method:



- **By Radiation Method:** This method is particularly suitable when a contoured map of a hill is required. A number of lines are set out, radiating at a given angular interval from each of the traverse stations, and the representative points on these lines are located in the field by observing, (i) The horizontal distances, and (ii) the Staff readings of the bottom, middle, and top wire.

The elevations and the distances of these points are then calculated. The survey is plotted and the contour lines are then interpolated.

By Tachometric / Radiation Method:



COMPARISON

S.NO	DIRECT METHOD	INDIRECT METHOD
1	The method is accurate but very slow & trouble some.	Not very accurate but quicker.
2	It is very expensive	It is very cheap
3	For small projects where greater accuracy is require.	For larger project where greater accuracy is not require.
4	It is unsuitable for hilly areas.	It is suitable for all type of areas.
5	Calculation is to be done in the field.	Calculation is not done in the field.